Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Menaker, Michael Ph.D.		
Project Title:	Circadian Disruption in the Space Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mm7e@virginia.edu	**Fax:**	FY 434-982-5626
PI Organization Type:	UNIVERSITY	**Phone:**	434-982-5767
Organization Name:	University of Virginia		
PI Address 1:	Department of Biology		
PI Address 2:	Gilmer Hall		
PI Web Page:			
City:	Charlottesville	**State:**	VA
Zip Code:	22904-4328	**Congressional District:**	5
Comments:	Deceased as of February 2021		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	03/01/2001	**End Date:**	06/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-HPF.002.09		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This proposal is directed at understanding the physiology of long-term circadian disruption (dysphasia) using a unique transgenic rat as a model system. This model enables us independently to assess rhythmicity of the master circadian oscillator in the brain, other brain oscillators, and the damped circadian oscillators that are found in almost every organ and tissue, and to infer whether normal synchrony is maintained in a variety of environments.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	In any long duration space flight rhythmicity or lack of it in the space environment will be of paramount concern to the health of the personnel. It is therefore important to learn what are the biological effects of the environments to which human space travelers are likely to be exposed. Some of these biological effects will be the result of disruption of circadian organization of the kind that produces the temporary malaise of "jet lag" and the more serious pathologies of long -term shift work. Circadian disruption in the space environment will be severe and long-lasting but it may be possible to mitigate its effects if we understand its physiology.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 03/07/2011)